

Lesson 13 More on Combinatorial Proofs

In this lesson, we give more examples of combinatorial proofs.

Proposition 1 Let n, k be non-negative integer with $0 \leq k \leq n$. Then

$$\binom{n}{k} = \binom{n}{n-k}$$

This identity has a straightforward algebraic proof using the formula for $\binom{n}{k}$:

$$\binom{n}{n-k} = \frac{n!}{(n-k)!(n-(n-k))!} = \frac{n!}{(n-k)!k!} = \binom{n}{k}$$

Now we present a combinatorial proof, which is more “satisfying” and intuitive.

Proof To construct a k -element subset of an n -element set, we may choose k elements to include or $n - k$ elements to *exclude*, i.e. the number of ways is $\binom{n}{k}$ or $\binom{n}{n-k}$, and the two must be the same.

Proposition 2 Let n, k be non-negative integer with $0 \leq k \leq n$. Then

$$k \binom{n}{k} = n \binom{n-1}{k-1}$$

Proof In a class of n students, how many ways are there to choose a committee of k members with a chair? There are two ways to answer this question.

LHS: First choose the committee, there are $\binom{n}{k}$ ways, and then choose the chair from the k committee members, there are k ways, so the answer is $k \binom{n}{k}$.

RHS: First choose the chair, there are n ways, then choose $k - 1$ committee members from the remaining $n - 1$ students, so the answer is $n \binom{n-1}{k-1}$.

Next, we look at examples where one of the two ways of counting involves breaking up into cases. The following proposition, called **Pascal’s identity**, is observed readily in Pascal’s triangle.

Proposition 3 (Pascal’s identity) Let $n, k \in \mathbb{N}$ with $0 \leq k \leq n$. Then

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$$

Proof The LHS is the number ways to choose k marbles from 1 red marbles and $n - 1$ blue marble. For the RHS, consider two ways of choosing k marbles:

- There are $\binom{1}{1} \binom{n-1}{k-1} = \binom{n-1}{k-1}$ ways to choose 1 red and $k - 1$ blue marbles.
- There are $\binom{n-1}{k}$ ways to choose all blue marbles.

Proposition 4 Let $n, m, l \in \mathbb{N}$, with $l \leq n + m$. Then

$$\sum_{k=0}^l \binom{n}{k} \binom{m}{l-k} = \binom{n+m}{l}$$

Proof RHS is the number of ways to choose l marbles from n red and m blue marbles, and we will show that the LHS is the same. There are $\binom{n}{0} \binom{m}{l}$ ways to choose no red and all blue marbles, $\binom{n}{1} \binom{m}{l-1}$ ways to choose subsets with one red and $l-1$ blue marbles, $\binom{n}{2} \binom{m}{l-2}$ ways to choose two red and $l-2$ blue marbles, etc. So, the total number of ways to choose n marbles from n red and m blue marbles is:

$$\binom{n}{0} \binom{m}{l} + \binom{n}{1} \binom{m}{l-1} + \binom{n}{2} \binom{m}{l-2} + \cdots + \binom{n}{l} \binom{m}{0} = \sum_{k=0}^l \binom{n}{k} \binom{m}{n-k}$$

Note that if $l > n$, then the $\binom{n}{k}$ terms where $k > n$ are simply zero.

Proposition 5 Let n be an integer greater than 1. Then

$$\binom{n}{2} = 1 + 2 + 3 + \cdots + (n-1)$$

Proof The LHS is the number 2-element subset from $\{1, 2, \dots, n\}$. The RHS gives a possible breakdown of the 2-element subsets.

- There are $n-1$ subsets with "1": $\{1,2\}, \{1,3\}, \dots, \{1,n\}$
- There are $n-2$ subsets without "1", but with "2": $\{2,3\}, \{2,4\}, \dots, \{2,n\}$
- There are $n-3$ subsets without "1" or "2", but with "3" : $\{3,4\}, \{3,5\}, \dots, \{3,n\}$
- $\vdots$
- There is 1 subset without any of "1", "2", ..., "n-2": $\{n-1, n\}$

In the last 3 identities, one side has a simple meaning: $\binom{n}{k}$ in Pascal's Identity, $\binom{n+m}{l}$ in Proposition 4, and $\binom{n}{2}$ in Proposition 5, while the other side is the sum of different cases which make up the simple side.